

GRAND ISLE, LA, USA

WMO: 994977

Lat: 29.2647N

Lon: 89.9576W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.2	4.5	-7.5	2.0	3.3	-4.4	2.6	5.9	14.4	8.0	12.9	7.9	8.3	10	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	32.1	25.8	31.4	25.6	30.8	25.5	26.8	29.6	26.4	29.3	26.2	29.1	2.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.3	28.5	25.6	20.9	28.5	25.3	20.4	28.5	83.6	29.5	82.2	29.3	81.0	29.2	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.0	9.3	0.2	33.8	2.0	1.8	-1.2	35.1	-2.4	36.2	-3.6	37.2	-5.0	38.6		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.7	12.8	14.5	17.8	21.3	25.0	27.8	28.5	28.6	27.5	23.5	18.3	14.8		
	DBStd	6.48	4.60	4.37	3.86	2.82	2.13	1.42	1.34	1.47	1.75	2.75	3.94	4.53		
	HDD10.0	57	27	12	3	0	0	0	0	0	0	0	2	13		
	HDD18.3	530	175	116	55	9	1	0	0	0	0	2	48	123		
	CDD10.0	4344	114	139	244	339	464	534	574	578	524	419	251	163		
	CDD18.3	1774	3	9	37	98	206	284	316	319	274	163	48	14		
	CDH23.3	17066	0	6	24	241	1536	3306	3947	4007	2985	937	77	1		
	CDH26.7	5351	0	0	0	2	150	1070	1546	1612	884	85	1	0		
(14) Wind	WSAvg	4.5	5.2	4.9	4.8	5.1	4.4	3.7	3.4	3.4	4.0	4.7	5.1	4.9		
(15) Precipitation	PrecAvg	1540	141	93	117	104	89	145	186	166	165	111	93	107		
	PrecMax	2289	470	258	326	536	199	381	358	304	397	326	364	178		
	PrecMin	1048	0	9	9	0	10	37	73	78	67	6	0	35		
	PrecStd	346	127	66	70	113	58	83	72	67	84	89	88	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	24.3	25.2	26.9	30.1	32.1	32.9	33.8	32.5	29.7	26.4	23.3		
		MCWB	20.4	20.8	21.5	22.8	23.2	25.3	25.2	27.9	25.5	25.8	24.6	21.9		
	2%	DB	20.6	22.3	24.1	26.1	28.8	31.3	31.8	32.4	31.0	28.4	25.1	22.3		
		MCWB	19.5	20.3	21.5	22.9	24.3	25.0	25.6	26.2	25.0	25.0	23.1	21.0		
	5%	DB	19.8	21.0	23.2	25.4	28.0	30.6	31.1	31.5	30.4	27.6	24.3	21.5		
		MCWB	18.8	19.6	20.9	22.6	24.2	24.8	25.6	25.8	24.9	24.9	22.6	20.3		
	10%	DB	18.9	20.0	22.4	24.8	27.5	30.1	30.6	30.8	29.8	26.9	23.4	20.6		
		MCWB	17.9	18.8	20.4	22.1	24.0	24.9	25.6	25.6	24.9	24.4	21.7	19.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.6	22.9	24.1	26.1	26.4	27.0	27.7	26.9	26.3	25.5	21.6		
		MCDB	20.9	21.2	23.4	25.2	27.5	28.8	29.6	31.6	28.8	27.4	26.5	22.1		
	2%	WB	19.6	19.9	22.0	23.6	25.4	26.0	26.7	26.8	26.4	25.7	24.3	20.8		
		MCDB	20.0	20.5	22.5	24.8	27.0	28.6	29.6	30.2	28.8	27.1	25.0	21.2		
	5%	WB	18.8	19.1	21.3	23.2	25.1	25.7	26.4	26.4	25.8	25.2	23.5	20.2		
		MCDB	19.3	19.7	22.0	24.5	26.8	28.5	29.4	29.6	28.5	26.7	24.4	20.7		
	10%	WB	17.9	18.3	20.5	22.6	24.7	25.4	26.1	26.2	25.5	24.6	22.5	19.2		
		MCDB	18.5	18.8	21.2	24.1	26.6	28.3	29.3	29.4	28.4	26.4	23.7	20.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	4.9	4.4	3.7	3.3	3.5	3.7	3.9	3.6	3.9	4.6	4.8		
		MCDBR	5.2	4.8	4.0	3.4	3.4	3.9	3.9	4.6	4.1	4.3	4.3	4.5		
	5% WB	MCWBR	4.9	4.0	3.5	2.7	1.9	2.4	2.3	2.5	2.2	1.9	4.0	4.7		
		MCWBR	5.2	5.2	3.8	2.9	2.8	3.4	3.7	3.7	3.4	3.1	4.4	4.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.348	0.363	0.377	0.420	0.444	0.463	0.492	0.492	0.444	0.390	0.364	0.353		
		taud	2.530	2.505	2.449	2.324	2.298	2.291	2.237	2.234	2.333	2.478	2.493	2.528		
	Ebn at Noon	Ebn at Noon	890	905	911	877	853	831	806	804	834	864	863	866		
		Edh at Noon	89	99	111	129	133	134	141	140	123	100	91	85		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.07	3.69	4.97	5.94	6.55	6.21	5.72	5.37	5.14	4.61	3.52	2.76		
	RadStd	RadStd	0.25	0.45	0.43	0.54	0.32	0.46	0.40	0.44	0.40	0.43	0.33	0.23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+106

Nomenclature: See separate page